

Regression problem example

Example: Let x and y equal the ACT scores in social science and natural science, respectively, for students applying for admission to a small liberal arts college. A sample of four students yielded the following data:

Students	x	y
A	32	28
B	23	25
C	23	24
D	30	27

We assume that the first score influences the second in nearly linear way.

- 1) Calculate the regression line for this data set.
- 2) Find the maximum likelihood estimator for the noise level (error with respect to the best line), if we model the noise as normal random variable $N(0, \sigma^2)$.
- 3) Find the forecast score in natural science for another student who got $x = 22$ in social science.

Linear regression line is

$$\mu(x) = \hat{\alpha} + \hat{\beta}(x - \bar{x})$$

$$\hat{\alpha} = \bar{y}$$
$$\hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Maximum likelihood estimate for σ^2 is

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta}(x_i - \bar{x}))^2$$

$\hat{\alpha}, \hat{\beta}, \bar{x}$ are constants computed from the available dataset

← computed in class (see section 6.5)

$$\text{Here, } \hat{\alpha} = \bar{y} = \frac{1}{4} (28 + 25 + 24 + 27) = 26$$
$$\bar{x} = \frac{1}{4} (32 + 23 + 23 + 30) = 27$$

($n=4$)

$$\hat{\beta} = \frac{28 \cdot (32-27) + 25 \cdot (23-27) + 24 \cdot (23-27) + 27 \cdot (30-27)}{(32-27)^2 + (23-27)^2 + (23-27)^2 + (30-27)^2}$$

$$= \frac{28 \cdot 5 + 25 \cdot (-4) + 24 \cdot (-4) + 27 \cdot 3}{5^2 + 4^2 + 4^2 + 3^2} = \frac{25}{66} \approx 0.38$$

~~1) So,~~ So, regression line is

$$1) \mu(x) = 26 + 0.38(x - 27) = \boxed{0.38x + 15.74}$$

(linear equation in x)

2) To find $\hat{\sigma}^2$, let us find all 4 residuals:

$$1st \text{ residual} = y_1 - \hat{\alpha} - \hat{\beta}(x_1 - \bar{x}) = y_1 - \mu(x_1) =$$

$$28 - 0.38 \cdot 32 - 15.74 = \underline{0.1}$$

$$2nd \text{ residual} = y_2 - \mu(x_2) = 25 - 0.38 \cdot 23 - 15.74 = \underline{0.52}$$

$$3rd \text{ residual} = y_3 - \mu(x_3) = 24 - 0.38 \cdot 23 - 15.74 = \underline{-0.48}$$

$$4th \text{ residual} = y_4 - \mu(x_4) = 27 - 0.38 \cdot 30 - 15.74 = \underline{-0.14}$$

We can see that residuals are small comparing to the data points

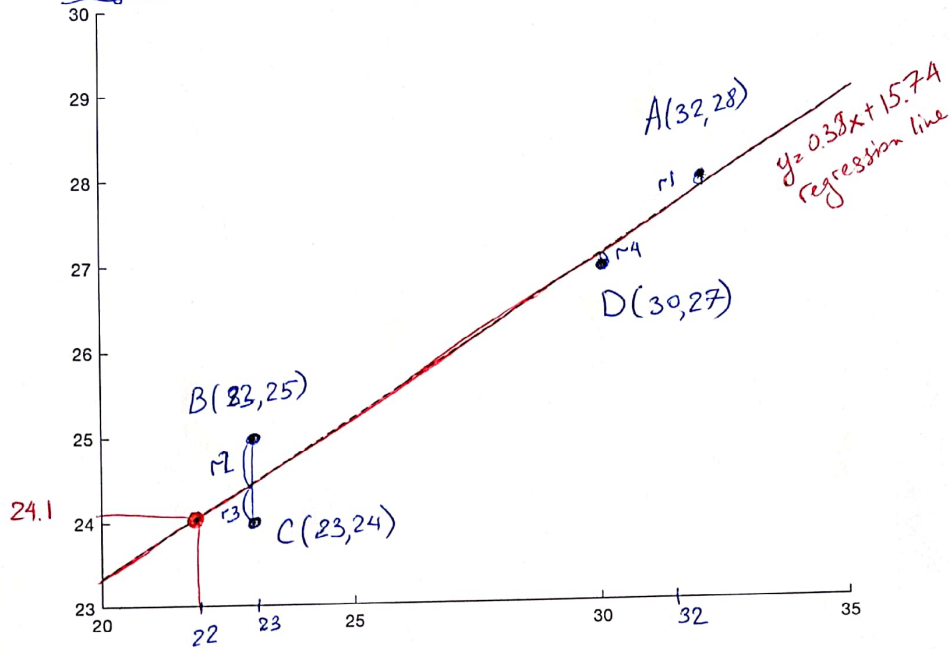
$$2) \hat{\sigma}^2 = \frac{1}{4} (0.1^2 + 0.52^2 + 0.48^2 + 0.14^2) = \frac{0.5304}{4} = \boxed{0.1326}$$

3) for $x=22$ forecast score

$$\hat{y} = \mu(x) = 0.38 \cdot 22 + 15.74 = 24.1$$

(or 24 if we assume that ACT scores are integers)

Regression line plot:



forecast for a student with $x=22$

• A, B, C, D - data points (x_i, y_i)

— Best fitting line (regression line)

$$r_1 = 0.1$$

$$r_2 = 0.52$$

$$r_3 = -0.48$$

$$r_4 = -0.14$$

} residuals

Plot of residuals:

